# HW 2: Agent-Based Modeling of Pandemic Spread

**Hypothesis:** Reducing agents' movement while holding other factors constant will lower the infection peak height and delay the peak time, with individual simulations varying slightly and the average curve broadly resembling the SIR model.

**The hypothesis is largely supported.** With reduced movement, the infection peak height is lower, and the timing is delayed. Individual runs vary slightly but generally resemble the average curve. The average agent-based curves broadly resemble the SIR curves. Infected agents rose and declined, while recovered agents increased. The susceptible curve did not increase slightly as hypothesized and stayed constant.

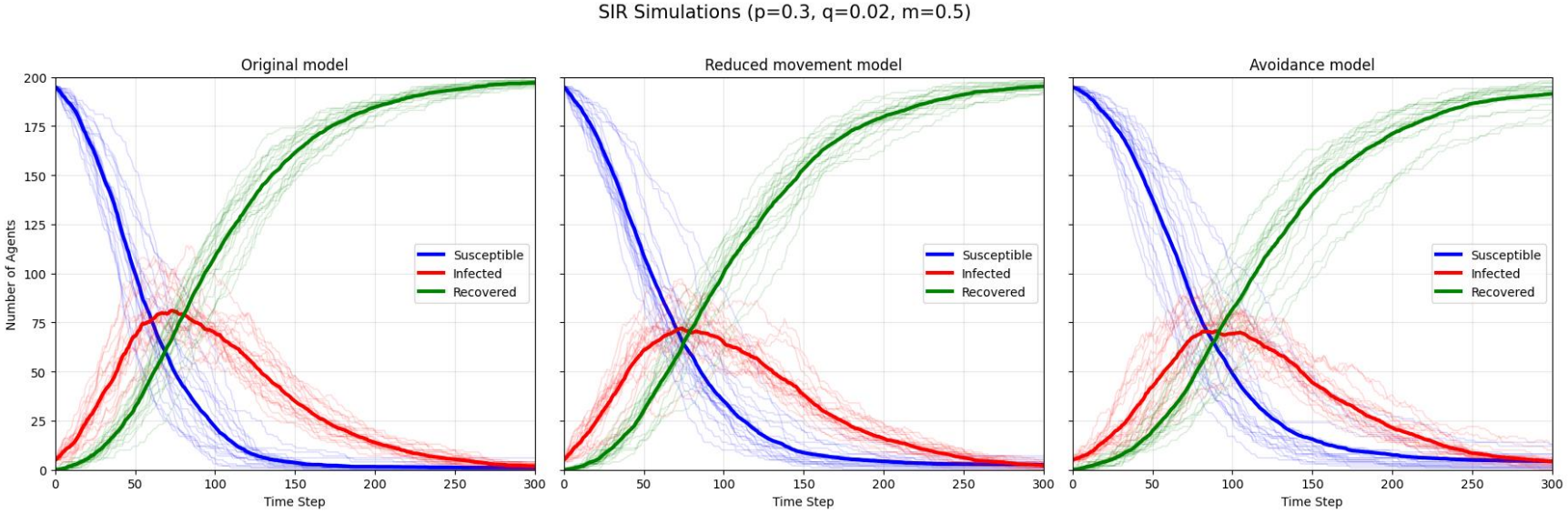



**Table 1:** Mean infection peak heights (agents) and time to infection peak (steps) of original, reduced movement, and avoidance model.

| | Mean peak height (agents) | Mean time to peak (steps) |
|---|---|---|
| Original | 89.10 | 71.60 |
| Reduced | 78.95 (↓) | 73.85 (↑) |
| Avoidance | 79.10 (↓) | 92.75 (↑) |

**Figure 1:** SIR curve of the original, reduced movement, and avoidance model (p = 0.3, q = 0.02). m = 0.5 for the reduced movement and avoidance model. p denotes the infection probability, q denotes the recovery probability, and m denotes the movement probability.

**Lesson learned from modeling:** It's important to define model parameters before modeling and record the number of each agent at each time step to facilitate future analysis/interpretation (e.g., finding time until peak).